

Structure, morphology and optical characterization of Dy³⁺-doped BaYF₅ nanocrystals for warm white light emitting devices

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ABSTRACT

The barium yttrium fluoride BaYF₅ nanocrystalline powders doped with different concentrations of Dy³⁺ ions have been synthesized via a hydrothermal method and studied their structural, morphological, thermal, vibrational, and optical properties. These nanopowders have been crystallized in a single phase of the tetragonal structure with the average size of around 30 nm having spherical shape in morphology. Upon excitations at 350 and 387 nm, Dy³⁺-doped BaYF₅ nanocrystals exhibit strong blue and yellow emissions ascribed to the ⁴F_{9/2} → ⁶H_{15/2} and ⁴F_{9/2} → ⁶H_{13/2} transitions, respectively. Decay curves of the ⁴F_{9/2} level of Dy³⁺ ion in BaYF₅ nanocrystals exhibit non-exponential nature due to the dipole-dipole interaction between Dy³⁺ ions, confirmed by Inokuti-Hirayama model. The quantum yield for these nanocrystals have been found to be increased from 4.64% to 11.61% as the concentration of Dy³⁺ ions increases from 1.0 mol% to 2.0 mol% and then decreased to 10.68% as the dopant concentration increased to 5.0 mol%. Moreover, color coordinates and correlated color temperatures have been evaluated as a function of concentration and excitation wavelength and found to be in the warm white light region for all Dy³⁺ concentrations.

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1. Introduction

Recently, nanoscale materials have attracted much attention since its physical, chemical, and electronic properties have been influenced significantly by their structure, size, morphology, and impurities [1–11]. Trivalent rare earth (RE³⁺) doped inorganic nanocrystals can be extensively used as nano-phosphors in display devices, fluorescent tubes, white light emitting devices, biological imaging, optical sensors, scintillators, etc., due to their superior thermal and chemical stability, and the unique optical properties [1–12]. Of these applications, white light emitting devices (WLEDs) are considered to be the next (fourth) generation light sources that will substitute current conventional incandescent and fluorescent

lamps, because of their energy saving capability, greater reliability, high efficiency, eco-friendly degradation, brightness and long operation time [3,11,12].

White light can be generated by three ways: First, by combining a red, a green, and a blue LEDs, although it can be quite costly and complex because each LED degrades at a different rate [13]; second, by exciting blue, green and red emitting phosphors with a UV LED excitation, but the strong re-absorption of the blue light by the green and red phosphors may significantly lower efficiency of the device [12]; and third, by yellow emitting Ce³⁺ doped Y₃Al₅O₁₂ phosphors under GaN-based blue LED excitation, although it suffers from chromatic aberration and poor white light performance after certain period of working due to their individual degradation and lack of red component [11]. Hence, it is interesting to search for intense white light emitting phosphors under UV/blue light LED excitation [11,12,14].

Among RE³⁺ ions, Dy³⁺ ion is a suitable activator for generation

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of white light due to its characteristic blue and yellow emissions. Therefore, it is possible to obtain white light emission from Dy^{3+} -doped optical materials by changing the ligands surrounding Dy^{3+} ions, the active ion concentration, the excitation wavelength or the laser pump power, which leads to tuning the intensity ratio of yellow to blue (Y/B) emissions [11,15].

It is well known that fluorides are efficient host matrices for luminescence due to their low phonon energies and wide optical (UV–NIR) transparency [2–5]. The BaYF_5 nanocrystals doped with RE^{3+} ions are considered to be a good optical material for Stokes and anti-Stokes luminescence [1–5,7,16]. Huang et al. [1] studied that the influences of experimental parameters like pH on crystal structure, size and morphology of BaYF_5 nanocrystals with average crystallite size about 24 nm, synthesized by hydrothermal method. Qiu et al. [2] observed the up-conversion emission in $\text{RE}^{3+}/\text{Yb}^{3+}$ ($\text{RE} = \text{Er}$, Ho and Tm)-doped tetragonal structured BaYF_5 nanocrystals with average crystallite size of 50 nm, synthesized via a hydrothermal method by controlling EDTA and pH values. Zhang et al. [3] synthesized the $\text{Er}^{3+}/\text{Tm}^{3+}/\text{Yb}^{3+}$ -doped BaYF_5 nanocrystals through the hydrothermal method and found that the nanocrystals have a tetragonal structure with average crystallite size about 18 nm and studied the white luminescence properties through up-conversion as a function of Yb^{3+} ion concentration. Lei et al. [4] prepared the hydrophilic $\text{Tb}^{3+}/\text{Ce}^{3+}$ -doped BaYF_5 nanocrystals in tetragonal structure with average crystallite size of 12 nm by microwave-assisted route and observed bright green fluorescence emission. Zhai et al. [5] prepared the $\text{Er}^{3+}/\text{Yb}^{3+}$ -doped BaYF_5 core-shell nanoparticles by a high boiling solvent process and found the synthesized particles are in tetragonal structure with the average size of 3 nm. Liu et al. [7] synthesized the $\text{Er}^{3+}/\text{Yb}^{3+}$: BaYF_5 nanoparticles in cubic structure with average crystallite size of 24 nm by a facile hydrothermal method and studied their up-conversion fluorescence and *in vivo* computed X-ray tomography bio-imaging. Cao et al. [16] prepared 2.0 mol% Dy^{3+} -doped BaYF_5 nanocrystals by hydrothermal method with tetragonal structure with the average size of 10 nm and studied temperature dependent luminescence for temperature sensor applications.

Hence, it is interesting to study structural, morphological, thermal, optical, and luminescence properties of Dy^{3+} -doped BaYF_5 nanocrystals prepared by the hydrothermal method for warm white light emitting device applications. Particle size and morphology of the synthesized nanocrystals have been studied by powder X-ray diffraction (XRD) and high-resolution transmission electron microscopy (HRTEM), respectively. The thermal stability of the synthesized nanocrystals has been studied by thermogravimetric (TG) analysis. The vibrational properties of BaYF_5 nanocrystals have been studied by using Fourier transmission infrared spectroscopy (FTIR). Diffuse reflectance spectrum has been measured to predict partial energy levels of Dy^{3+} ions in BaYF_5 nanocrystals. Excitation and emission spectra of the prepared samples have been measured as a function of Dy^{3+} ions concentration. The luminescent decay curves for the $^4\text{F}_{9/2}$ level of Dy^{3+} ions have been measured and analyzed by the Inokuti-Hirayama model to know the kind of multipolar interaction responsible for energy transfer among Dy^{3+} ions. Finally, the Commission International d'Eclairage (CIE) coordinates and Correlated Color Temperature (CCT) values are obtained for Dy^{3+} -doped BaYF_5 nanocrystals by varying the Dy^{3+} concentration and excitation wavelength.

2. Experimental details

2.1. Synthesis of Dy^{3+} -doped BaYF_5 nanocrystals

$\text{BaY}_{1-x}\text{Dy}_x\text{F}_5$ phosphors (where $x = 0.01, 0.02, 0.03, 0.04$, and

0.05, labelled as BYF1Dy, BYF2Dy, BYF3Dy, BYF4Dy and BYF5Dy) were prepared by a hydrothermal method. $\text{Dy}(\text{NO}_3)_3$ (Aldrich, 99.9%), $\text{Y}(\text{NO}_3)_3$ (Hi-media, 99.0%), $\text{Ba}(\text{NO}_3)_2$ (Aldrich, 99.999%), NH_4F (Aldrich, 99.99%), NH_4OH (Aldrich, 99.99%) and EDTA solution (Nice chemicals Ltd.) were used as starting reagents. Stoichiometric quantities of $\text{Ba}(\text{NO}_3)_2$, $\text{Dy}(\text{NO}_3)_3$, $\text{Y}(\text{NO}_3)_3$, and EDTA ($\text{C}_{10}\text{H}_{16}\text{N}_2\text{O}_8$) were dissolved in 20 ml of deionized water under stirring. The solution was regulated to a pH = 9 with NH_4OH solution and finally added 1 mmol of NH_4F . After vigorous stirring for 1 h, the mixture was transferred into a 50 mL PTFE-lined stainless steel autoclave and hydrothermally reacted at 180 °C for 12 h. The precipitation was collected by centrifugation, washed two times with distilled water and centrifuged for 10 min each time. Then, the precipitation was dried for 12 h at 80 °C and finally, the product was annealed at 600 °C for 2 h.

2.2. Characterization

Powder XRD patterns of the synthesized phosphors were measured using the $\text{CuK}_{\alpha 1}$ (1.5406 Å) radiation with a step size of 0.02° (RIGAKU; Miniflex-600). Scanning electron micrograph was measured using a LEO 440 PC digital scanning electron microscope (SEM) whereas high-resolution transmission electron micrographs were measured using a JEOL microscope 200 kV to know the morphology of synthesized samples. Thermogravimetric (TG) analysis was carried out to estimate the thermal stability of the sample. The vibrational modes of the BaYF_5 were studied using Fourier transform infrared spectra measured from 400 to 4000 cm^{-1} (Perkin–Elmer Rx1). The diffuse reflectance spectrum in UV–Vis–NIR range was measured with a spectrophotometer (Agilent Technologies Cary 5000). The photoluminescence excitation (PLE) and emission spectra were recorded by a spectrometer equipped with a 450 W xenon arc lamp (CW) as a light source (Edinburgh FLS980). The quantum yield measurements were carried out using steady state spectrometer (Edinburgh instrument FLSP920) with a xenon lamp as an excitation source and equipped with an integrating sphere. Luminescence decay curves of the $^4\text{F}_{9/2}$ level of Dy^{3+} ions were measured under 473 nm laser excitation using a 10 ns optical parametric oscillator (EKSPLA/NT342/3/UVE) and a digital storage oscilloscope (Lecroy WS424) coupled to a spectrometer (Jobin-Yvon TRIAX180) with a PMT (Hamamatsu R928) detection system. All the measurements were carried out at room temperature.

3. Results and discussion

3.1. Structure of Dy^{3+} -doped BaYF_5 nanocrystals

The crystal structure and phase purity of BaYF_5 : $x\text{Dy}^{3+}$ ($x = 1, 2, 3, 4$ and 5 mol%) powders have been studied by powder XRD technique (see Fig. 1 for BYF1Dy & BYF5Dy). Results reveal that all the reflections in the profile are well-indexed to a single phase tetragonal structure belonging to P-42₁m space group (No. 113, $Z = 10$) and the lattice parameters are found to be $a = b = 12.468$ Å and $c = 6.78$ Å. As can be seen from Fig. 1, peaks have slightly been shifted towards higher angle side with the increase of Dy^{3+} ions doping concentration into BaYF_5 nanocrystals. The average crystallite size (D) is determined by using Debye-Scherrer formula

$$D = \frac{k\lambda}{\beta \cos \theta} \Rightarrow \beta = \frac{k\lambda}{D \cos \theta} \quad (1)$$

where, λ is the wavelength of X-ray, β is full width at half maximum (FWHM); θ is the angle of diffraction peak, and k is the shape factor (0.89). The crystallite size of all Dy^{3+} -doped BaYF_5 nanocrystals is

evaluated and presented in Table 1. As can be seen from Table 1, the average crystallite size is nearly constant for all Dy^{3+} -doped BaYF_5 nanocrystals with a value of around 16 nm. Furthermore, lattice strain has been evaluated by using Williamson–Hall (W-H) method [17].

$$\beta = \frac{0.89\lambda}{D \cos \theta} + 4\varepsilon \tan \theta$$

$$\Rightarrow \beta \cos \theta = \frac{0.89\lambda}{D} + 4\varepsilon \sin \theta \quad (2)$$

where, ' ε ' is the lattice strain induced in crystalline powders. A plot of $\beta \cos \theta$ versus $4 \sin \theta$ is drawn to evaluate the lattice strain and crystallite size (see Fig. 2), and the values were tabulated (see Table 1). It is clearly observed that the crystallite size obtained from Scherrer's equation is smaller than that computed by W-H method. The similar results have been reported for $\text{Lu}_3\text{Ga}_5\text{O}_{12}$, ZnO and $\text{Fe}_2\text{Mn}_2\text{Ni}_{0.5}\text{Zn}_{1.5}\text{O}_9$ nanocrystals [18–20]. This is due to the fact that the Scherrer's equation could not take into the account of lattice strain on the peak broadening [18].

The morphology of these nanocrystals has been studied by scanning electron microscopy (SEM) and high-resolution transmission microscopy (HRTEM) and their respective micrographs have been shown in Fig. 3(a) and (b) for 4.0 mol% of Dy^{3+} -doped BaYF_5 nanocrystals. As can be seen, nanocrystals have been agglomerated in the form of interconnected spheres (see inset of Fig. 3(b)) with the narrow size distribution of around 25–40 nm. The average size of these nanocrystals is well agreed with the crystallite size obtained from XRD technique. Similar results are observed for other synthesized samples.

In order to investigate the change in mass and thermal stability of the Dy^{3+} -doped BaYF_5 nanocrystals, the TG–DTA was carried out (see Fig. 4). As can be seen from the TG curve, one can observe that weight of BaYF_5 nanocrystals does not change with the increase of temperature up to at about 650 °C indicating that the synthesized BaYF_5 nanocrystals are thermally stable. It can also be observed that weight has been drastically reduced beyond 650 °C temperature, which may correspond to dehydration of BaYF_5 and/or to the incorporation of oxygen in the crystal structure by replacing the

Table 1

Crystallite size and lattice strain for the Dy^{3+} -doped BaYF_5 nanocrystals along with other nanocrystalline materials.

Sample	Crystallite size (nm)		Lattice strain (%) Ref.	
	Debye-Scherrer	Williamson-Hall		
BYF1Dy	16	50	0.23	Present
BYF2Dy	17	40	0.16	Present
BYF3Dy	16	39	0.18	Present
BYF4Dy	17	39	0.15	Present
BYF5Dy	16	38	0.14	Present
LuGG1Sm	37	69	0.22	[18]
ZnO	27	35	0.13	[19]
$\text{Fe}_2\text{Mn}_2\text{Ni}_{0.5}\text{Zn}_{1.5}\text{O}_9$	17	28	0.40	[20]

fluorides, generating an oxyfluoride structure at higher temperatures. When oxygen replaces fluoride, the molecular weight decreases and hence, the weight of the product. On the other hand, the DTA curve does not found any clear phase change indicating that the synthesized BaYF_5 nanocrystals are thermally stable.

FTIR spectrum of 4.0 mol% of Dy^{3+} -doped BaYF_5 nanocrystals is shown in Fig. 5. The peak at around 470 cm^{-1} is attributed to Ba–F or Y–F bending mode, which is the characteristic of phonon energy of the BaYF_5 host matrix. Peaks at around 1100 and 1380 cm^{-1} are may be due to symmetric stretching vibration of C–O and N–O groups, respectively, whereas those located at 1480 and 1620 cm^{-1} are associated with the asymmetric and symmetric stretching vibrations of carboxylate (C=O) anions bound to the BaYF_5 nanocrystal surface [5]. The strong absorptions at 2850 and 2920 cm^{-1} are due to the symmetric and asymmetric stretching of CH_2 in the chain of EDTA, respectively. The broad peak centered around 3450 cm^{-1} corresponds to the O–H symmetric stretching from the surface hydroxyl group [21]. The similar results have also been observed in $\text{Er}^{3+}/\text{Yb}^{3+}$ -doped BaYF_5 nanoparticles and $\text{Ho}^{3+}/\text{Yb}^{3+}$ -doped BaGdF_5 nanoparticles [5,22]. These hydroxyl groups are responsible for the observed quenching of emission intensity of Dy^{3+} ions in BaYF_5 nanoparticles, as described in coming sections.

3.2. Diffuse reflectance and photoluminescence excitation spectra

The diffuse reflectance spectrum for 5.0 mol% Dy^{3+} ions -doped BaYF_5 powders have been measured in 300–1800 nm range and are shown in Fig. 6. Peaks correspond to intra-configurational $4f^9-4f^9$ electronic transitions arising from the $^6\text{H}_{15/2}$ ground state to the different excited levels of the Dy^{3+} ion and have been assigned by using the well-known Dieke's diagram for Dy^{3+} ion in LaCl_3 [23]. Fig. 6 shows that the diffuse reflectance spectrum of Dy^{3+} -doped BaYF_5 powder consists of twelve absorption peaks which are centered at around 324, 350, 364, 387, 451, 472, 745, 802, 903, 1097, 1283, and 1711 nm corresponding to the $^6\text{H}_{15/2} \rightarrow ^6\text{P}_{3/2}$, $^6\text{P}_{7/2}$, $^4\text{I}_{11/2}$, $^4\text{I}_{13/2}$, $^4\text{I}_{15/2}$, $^4\text{F}_{9/2}$, ($^6\text{F}_{1/2}$, $^6\text{F}_{3/2}$), $^6\text{F}_{5/2}$, $^6\text{F}_{7/2}$, ($^6\text{H}_{7/2}$, $^6\text{F}_{9/2}$), ($^6\text{F}_{11/2}$, $^6\text{H}_{9/2}$), and $^6\text{H}_{11/2}$ transitions, respectively. A partial energy level diagram of the Dy^{3+} ion in the BaYF_5 nanocrystals is shown in Fig. 7, which illustrates the excitation, de-excitation and cross-relaxation channels of Dy^{3+} ions in BaYF_5 nanocrystals.

For a better description of those bands observed below 400 nm, photoluminescence excitation (PLE) spectra have been obtained for the Dy^{3+} -doped BaYF_5 nanocrystals monitoring the well-known emission at 580 nm (see Fig. 8). The seven peaks observed at 324, 350, 364, 387, 428, 451, and 472 nm can be attributed to the $^6\text{H}_{15/2} \rightarrow ^6\text{P}_{3/2}$, $^6\text{P}_{7/2}$, $^4\text{I}_{11/2}$, $^4\text{I}_{13/2}$, $^4\text{G}_{11/2}$, $^4\text{I}_{15/2}$, and $^4\text{F}_{9/2}$ transitions, respectively. It is worth noting the large contribution of the $^6\text{P}_{7/2}$, $^4\text{I}_{11/2}$ and $^4\text{I}_{13/2}$ multiplets to the emission of the $^4\text{F}_{9/2}$ level. In addition, the intensity of the peaks increases up to 2.0 mol% Dy^{3+} -doping and then gradually decreases.

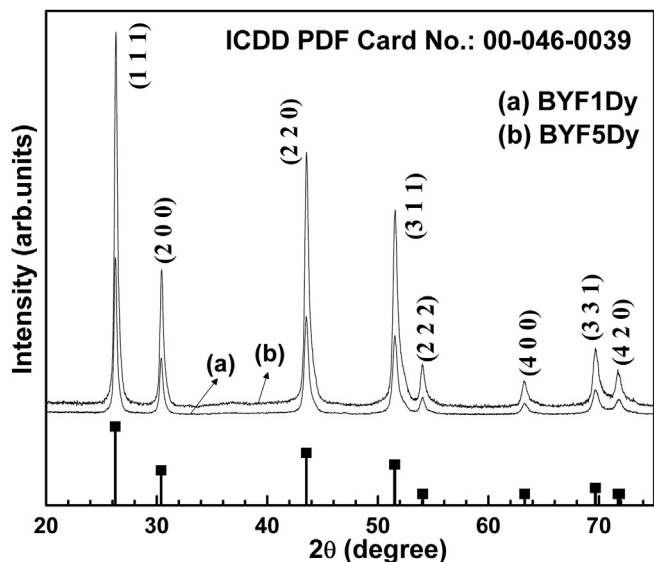


Fig. 1. XRD pattern of (a) 1.0 mol% and (b) 5.0 mol% Dy^{3+} -doped BaYF_5 nanocrystals. The vertical marks are the allowed reflections for this material in the $\text{P-42}_1\text{m}$ (No.113, $Z = 10$) space group.

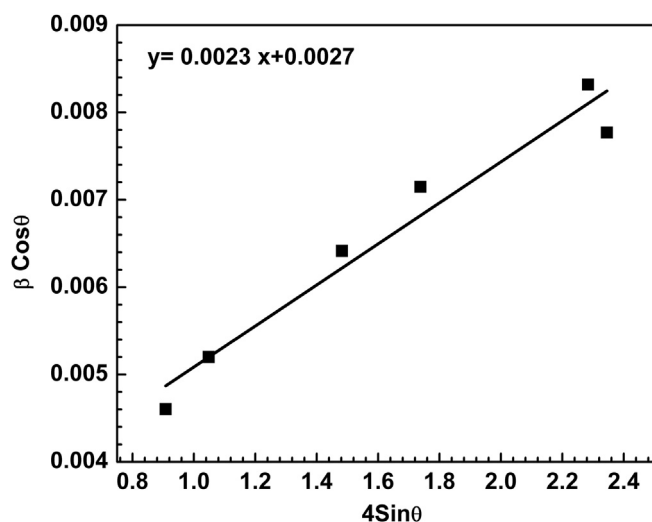


Fig. 2. The Williamson-Hall plot for 1.0 mol% Dy^{3+} -doped BaYF_5 nanocrystals.

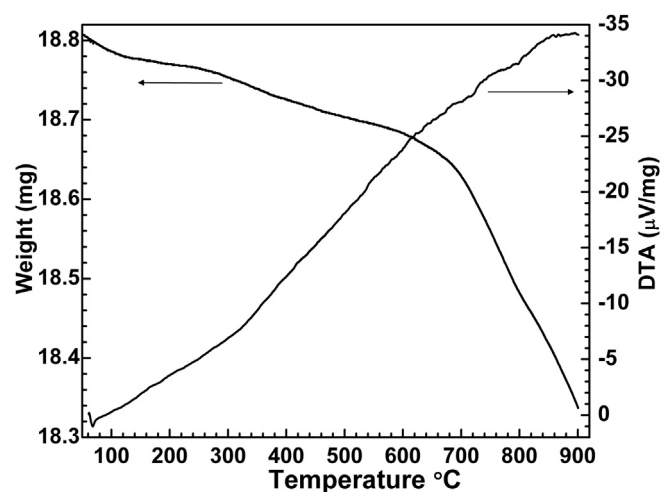


Fig. 4. TG-DTA curves of Dy^{3+} -doped BaYF_5 nanocrystals.

3.3. Luminescence spectra

In order to investigate the luminescent properties of Dy^{3+} ions in BaYF_5 nanocrystals, the emission spectra have been studied under resonant excitations of the ${}^6\text{H}_{15/2} \rightarrow {}^6\text{P}_{7/2}$ transition at 350 nm and the ${}^6\text{H}_{15/2} \rightarrow {}^4\text{I}_{13/2}$ transition at 387 nm [see Fig. 9(a) and (b)]. The characteristic emission from Dy^{3+} ions is observed at around 480, 570 and 660 nm, ascribed to the ${}^4\text{F}_{9/2} \rightarrow {}^6\text{H}_j$ ($j = 15/2, 13/2$ and $11/2$) transitions, respectively. The yellow emission corresponds to the hypersensitive ${}^4\text{F}_{9/2} \rightarrow {}^6\text{H}_{13/2}$ transition ($\Delta L = 2$; $\Delta J = 2$) and its intensity is highly influenced by the environment around the Dy^{3+} ion in the host lattice. When Dy^{3+} is located in low symmetry sites, the yellow emission is often prominent, but if located at sites with inversion center, the yellow emission becomes zero since electric-dipole transitions are forbidden in those sites [11]. The inset of Fig. 9(a) and (b) reveals that the intensity of the ${}^4\text{F}_{9/2} \rightarrow {}^6\text{H}_{13/2}$ (yellow) transition is relatively higher than that of the ${}^4\text{F}_{9/2} \rightarrow {}^6\text{H}_{15/2}$ (blue) transition, indicating that Dy^{3+} ions occupy lattice sites without inversion symmetry since Dy^{3+} -doped BaYF_5 nanocrystals have tetragonal symmetry, as evidenced from XRD.

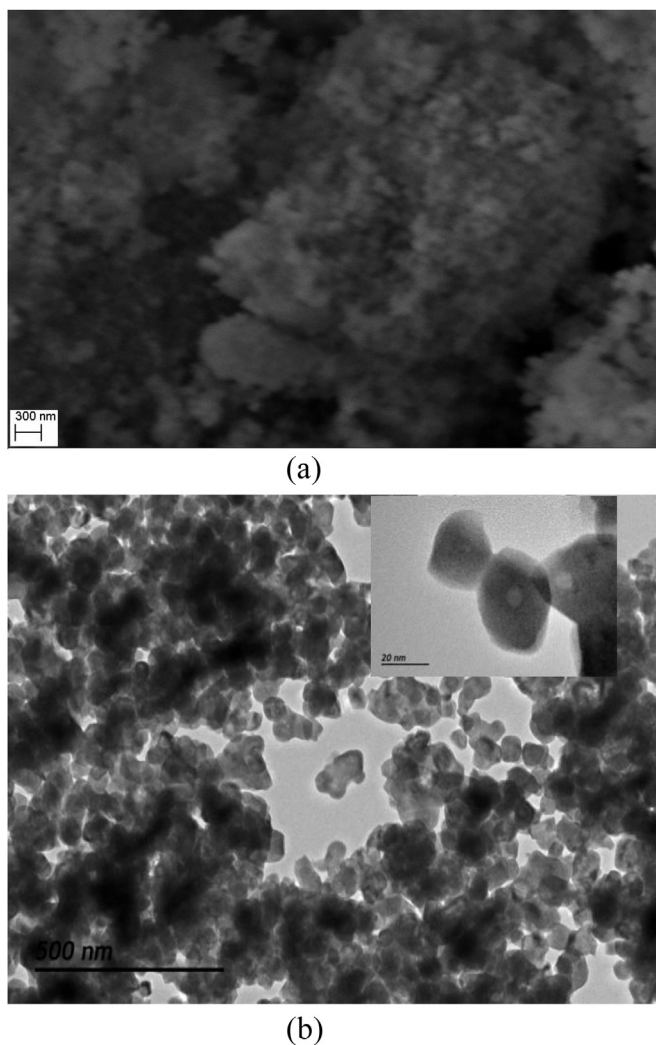


Fig. 3. (a) SEM micrograph and (b) HRTEM micrograph of 4.0 mol% Dy^{3+} -doped BaYF_5 nanocrystals.

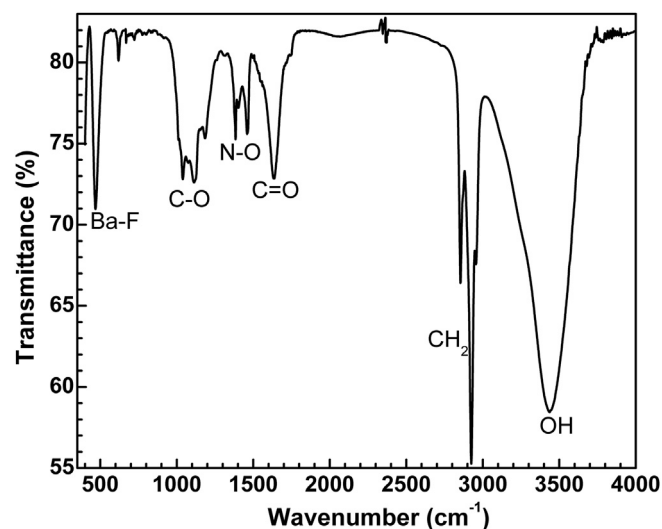


Fig. 5. Fourier transform infrared spectrum of 4.0 mol% Dy^{3+} -doped BaYF_5 nanocrystals.

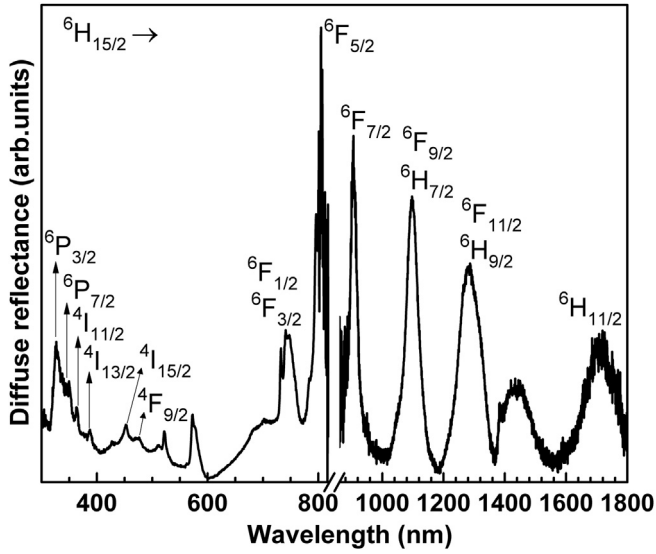


Fig. 6. Diffuse reflectance spectra of 5.0 mol% Dy^{3+} -doped BaYF_5 nanocrystals.

It is clearly observed from Fig. 9(a) and (b) that there is no change in peak positions under 350 and 387 nm excitation in all Dy^{3+} -doped BaYF_5 nanocrystals. Moreover, for the same experimental conditions the emission intensities of Dy^{3+} ions are found to increase from 1.0 to 2.0 mol% and then decreases due to concentration quenching through non-radiative energy transfer processes among Dy^{3+} ions (see inset of Fig. 9(a) and (b)). This indicates that the optimum molar concentration of Dy^{3+} ions is about 2.0 mol%. Similar results have been observed in Dy^{3+} -doped $\text{Lu}_3\text{Ga}_5\text{O}_{12}$, NaYF_4 , $\text{Gd}_3\text{Ga}_5\text{O}_{12}$, and GdF_3 [11,24–26]. Furthermore, the yellow to blue (Y/B) ratio as a function of active ion concentration have been calculated under 350 nm excitation and are found to be 1.98, 1.49, 2.07, 1.61, and 2.11 for BYF1Dy, BYF2Dy, BYF3Dy, BYF4Dy and BYF5Dy nanocrystals, respectively. Similar results were reported in Dy^{3+} -doped Y_2CaZnO_5 nanophosphor [27].

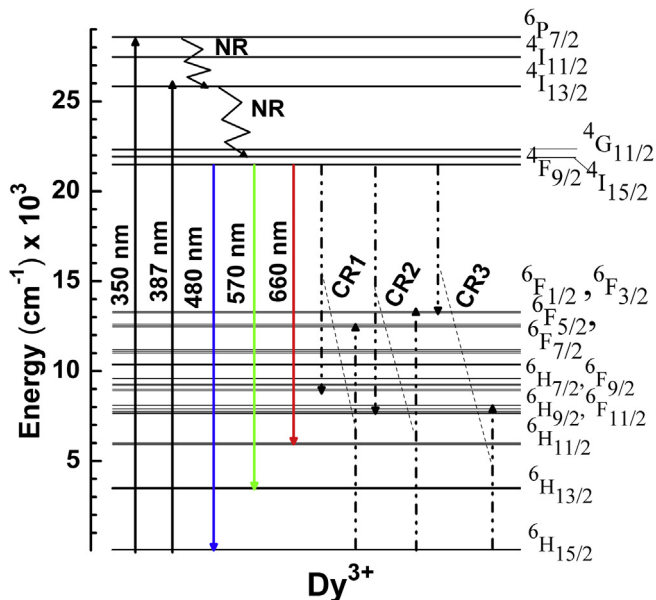


Fig. 7. Partial energy level structure of Dy^{3+} ions in BaYF_5 nanocrystals showing excitation, emission and cross-relaxation channels.

3.4. Decay curves

The luminescence decay curves of the $^4\text{F}_{9/2}$ emitting level of Dy^{3+} ion in BaYF_5 nanocrystals have been obtained under 473 nm excitation, in resonance with the $^6\text{H}_{15/2} \rightarrow ^4\text{F}_{9/2}$ absorption, by monitoring the $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{13/2}$ emission at 580 nm (see Fig. 10). The decay curves exhibit non-exponential nature for all Dy^{3+} concentrations in BaYF_5 nanocrystals (see Fig. 10). Hence, the effective lifetime (τ_{eff}) can be evaluated by the following equation:

$$\tau_{\text{eff}} = \frac{\int tI(t)dt}{\int I(t)dt} \quad (3)$$

where, $I(t)$ is the intensity of the luminescence at time t .

The effective lifetime of the $^4\text{F}_{9/2}$ level of Dy^{3+} ions is found to be 1.3, 1.21, 1.18, 1.01 and 0.97 ms for 1.0, 2.0, 3.0, 4.0, and 5.0 mol% of Dy^{3+} -doped BaYF_5 nanocrystals, respectively. It is clearly observed that the effective lifetime of the $^4\text{F}_{9/2}$ level is decreased with the increase of Dy^{3+} concentration (see Fig. 11). Similar results were reported for Dy^{3+} -doped BaGdF_5 , $\alpha\text{-NaYF}_4$, $\beta\text{-NaYF}_4$, $\text{Gd}_3\text{Ga}_5\text{O}_{12}$, Y_2CaZnO_5 , KY_3F_{10} , LiLuF_4 , Y_2O_3 , BaY_2F_8 , CaWO_4 , $\text{Y}_3\text{Ga}_5\text{O}_{12}$, $\text{Li}_4\text{CaB}_2\text{O}_6$, $\text{NaGd}(\text{MoO}_4)_2$, Gd_2O_3 , $\text{KLa}(\text{PO}_3)_4$ and K_2GdF_5 [10,24,25,27–37] and are shown in Table 2. It is also found from Table 2 that the effective lifetime of the $^4\text{F}_{9/2}$ level of Dy^{3+} -doped BaYF_5 nanocrystals is higher compared to Dy^{3+} -doped NaYF_4 , BaY_2F_8 , KY_3F_{10} , LiLuF_4 , Y_2O_3 , $\text{Gd}_3\text{Ga}_5\text{O}_{12}$, CaWO_4 , $\text{Y}_3\text{Ga}_5\text{O}_{12}$, LuNbO_4 , $\text{NaGd}(\text{MoO}_4)_2$ and Y_2CaZnO_5 and low compared to BaGdF_5 , K_2GdF_5 , and $\text{Li}_4\text{CaB}_2\text{O}_6$ phosphors. Similar concentration (1.0 mol %) of Dy^{3+} ions in different hosts have been chosen to plot the variation of lifetimes of $^4\text{F}_{9/2}$ level as a function of phonon energy. It is observed that the lifetime of the $^4\text{F}_{9/2}$ level of Dy^{3+} ion decreases with increase in phonon energy of the host [38–42] except in LiLuF_4 and Y_2CaZnO_5 phosphor, shown in Fig. 12.

The enhancement of non-exponential nature and shortening of the decay curves with an increase in the Dy^{3+} ion concentration is attributed to energy transfer from the Dy^{3+} ion in an excited $^4\text{F}_{9/2}$ state to a nearby Dy^{3+} ion in the ground $^6\text{H}_{15/2}$ state (see Fig. 10). The following resonant or quasi-resonant cross-relaxation channels are responsible for this energy transfer process [11] (see Fig. 7),

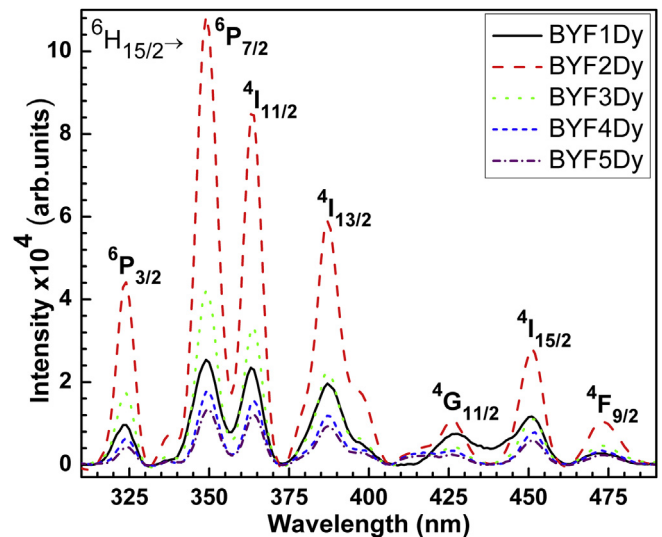


Fig. 8. Excitation spectra of Dy^{3+} -doped BaYF_5 nanocrystals monitoring the emission at 580 nm.

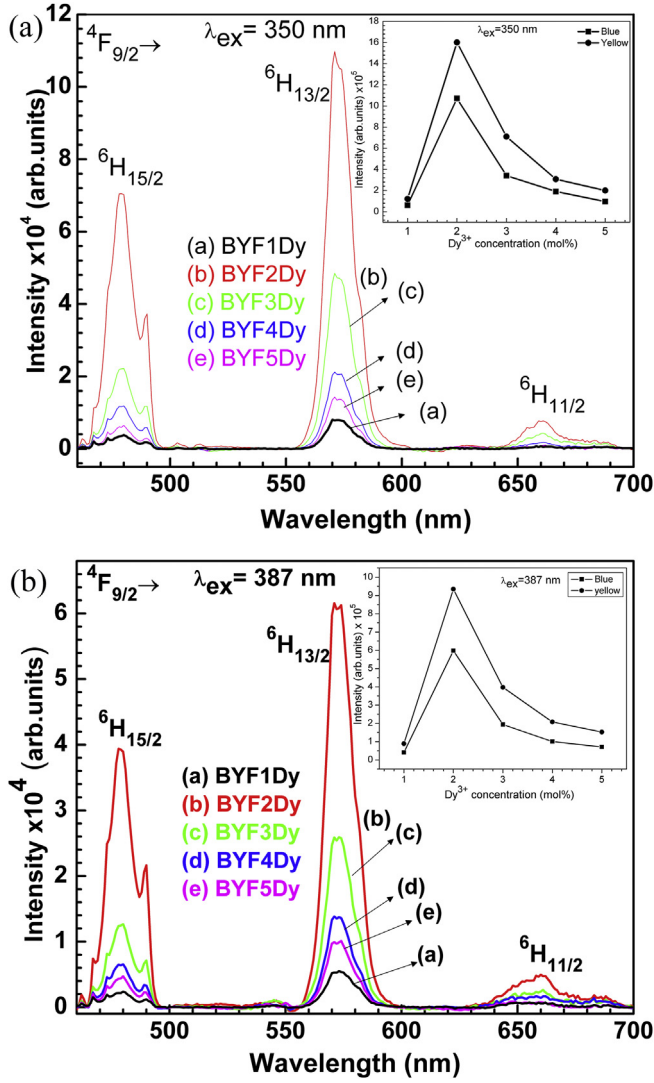


Fig. 9. (a). Luminescence spectra of Dy^{3+} -doped BaYF_5 nanocrystals obtained for different concentrations exciting at 350 nm. The inset shows the variation of blue (■) and yellow (●) emissions as a function of Dy^{3+} ion concentration. (b). Luminescence spectra of Dy^{3+} -doped BaYF_5 nanocrystals obtained for different concentrations exciting at 387 nm. The inset shows the variation of blue (■) and yellow (●) emissions as a function of Dy^{3+} ion concentration. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$\text{CR1: } ({}^4\text{F}_{9/2}, {}^6\text{H}_{15/2}) \rightarrow ({}^6\text{F}_{9/2} + {}^6\text{H}_{7/2}, {}^6\text{F}_{5/2})$$

$$\text{CR2: } ({}^4\text{F}_{9/2}, {}^6\text{H}_{15/2}) \rightarrow ({}^6\text{F}_{11/2} + {}^6\text{H}_{9/2}, {}^6\text{F}_{3/2} + {}^6\text{F}_{1/2})$$

$$\text{CR3: } ({}^4\text{F}_{9/2}, {}^6\text{H}_{15/2}) \rightarrow ({}^6\text{F}_{3/2} + {}^6\text{F}_{1/2}, {}^6\text{F}_{11/2} + {}^6\text{H}_{9/2})$$

In order to know the dominant mechanism of multipolar interaction responsible for energy transfer between the Dy^{3+} ions, the non-exponential decay curves have been analyzed by Inokuti-Hirayama model [43]. According to this model, luminescence intensity $I(t)$ is described as,

$$I(t) = I_0 \exp\left(-\frac{t}{\tau_0} - Q\left(\frac{t}{\tau_0}\right)^{3/S}\right) \quad (4)$$

where, I_0 is the emission intensity at $t = 0$; τ_0 is the intrinsic lifetime; S is the multipolar interaction parameter [$S = 6$

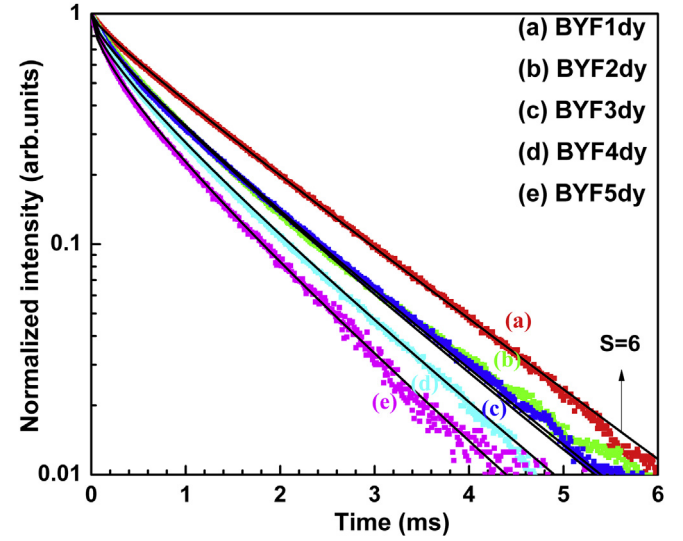


Fig. 10. Luminescence decay curves of the ${}^4\text{F}_{9/2}$ level as a function of the Dy^{3+} ion concentration in BaYF_5 nanocrystals. The solid lines (in black color) correspond to the IH fitting for $S = 6$.

(dipole–dipole), 8 (dipole–quadrupole) and 10 (quadrupole–quadrupole)]; and Q is the energy transfer parameter and is described by

$$Q = \frac{4\pi}{3} \Gamma \left[1 - \frac{3}{S}\right] N_0 R_0^3 \quad (5)$$

where N_0 is the total concentration of Dy^{3+} ions; and R_0 is the critical transfer distance between dopants.

All the decay curves are well fitted to Eq. (4) and the best fits are obtained for $S = 6$ for BYF1Dy, BYF2Dy, BYF3Dy, BYF4Dy and BYF5Dy, taking an intrinsic lifetime of 1.53 ms (is obtained by varying τ_0 and Q in the fitting process), indicating that the dominant mechanism of interaction between Dy^{3+} ions is of dipole–dipole type. The solid lines (in black color) in Fig. 10 correspond to the Inokuti-Hirayama model fitting for $S = 6$. The similar results have also been observed in Dy^{3+} -doped BaGdF_5 , BaY_2F_8 , $\text{Y}_3\text{Ga}_5\text{O}_{12}$, $\text{KLa}(\text{PO}_3)_4$ and K_2GdF_5 phosphors [10,30,32,36,37]. The energy

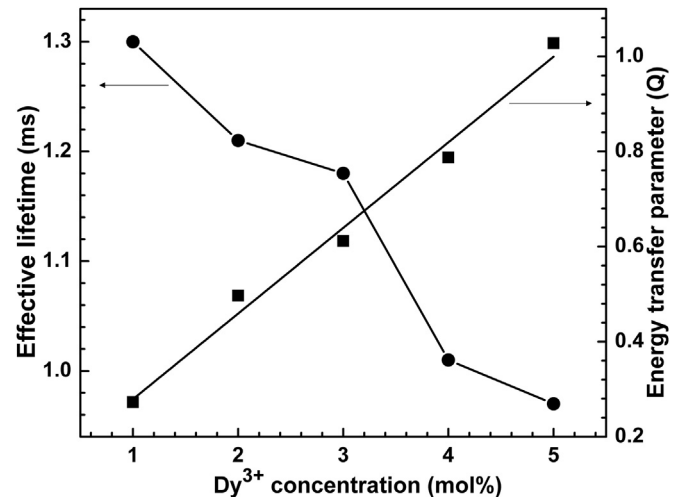


Fig. 11. Variation of effective lifetime obtained from Eq. (3) (●) and energy transfer parameter (■) as a function of Dy^{3+} concentration in BaYF_5 nanocrystals.

Table 2Comparison of lifetime of $^4F_{9/2}$ level of Dy^{3+} ions, multipolar interactions and their phonon energy in different crystalline materials.

Sample	λ_{ex} (nm)	τ_{eff} (ms)	Multipolar interactions	Phonon energy (cm^{-1})	Ref.
BYF1Dy (nano)	473	1.30	D-D	470	Present
BYF2Dy (nano)	473	1.21	D-D		Present
BYF3Dy (nano)	473	1.18	D-D		Present
BYF4Dy (nano)	473	1.01	D-D		Present
BYF5Dy (nano)	473	0.97	D-D		Present
BaGdF ₅ : 3%Dy ³⁺ (nano)	384	1.689	D-D		[10]
α -NaYF ₄ : 2%Dy ³⁺ (bulk)	350	0.396	—		[24]
β -NaYF ₄ : 2%Dy ³⁺ (bulk)	351	0.369	—		[24]
Gd ₃ Ga ₅ O ₁₂ : 2%Dy ³⁺ (nano)	352	0.876	—		[25]
Y ₂ CaZnO ₅ : 1%Dy ³⁺ (nano)	424	0.479	—	872	[27]
KY ₃ F ₁₀ : 1%Dy ³⁺ (bulk)	355	0.440	—	495	[28]
LiLuF ₄ : 1%Dy ³⁺ (bulk)	355	0.582	—	446	[28]
Y ₂ O ₃ : 1%Dy ³⁺ (nano)	355	0.305	—	597	[29]
BaY ₂ F ₈ : 0.5%Dy ³⁺ (bulk)	355	1.24	D-D		[30]
CaWO ₄ : 1%Dy ³⁺ (bulk)	353	0.153	—	912	[31]
Y ₃ Ga ₅ O ₁₂ : 1%Dy ³⁺ (bulk)	366	0.001	D-D	750	[32]
Li ₄ CaB ₂ O ₆ : 1%Dy ³⁺ (nano)	349	1.49	—		[33]
NaGd(MoO ₄) ₂ : 1.85%Dy ³⁺ (bulk)	454	0.156	—		[34]
Gd ₂ O ₃ : 1%Dy ³⁺ (nano)	238	0.34	—	600	[35]
KLa(PO ₃) ₄ : 1%Dy ³⁺ (bulk)	325	0.82	D-D		[36]
K ₂ GdF ₅ : 5% Dy ³⁺ (bulk)	—	1.14	D-D		[37]

D-D: Dipole-Dipole.

transfer parameter Q , obtained from the fitting, increases linearly with the increase of Dy^{3+} ion concentration in accordance with Eq. (5) as shown in Fig. 11.

3.5. Quantum yield

The absolute quantum yield of the Dy^{3+} -doped BaYF₅

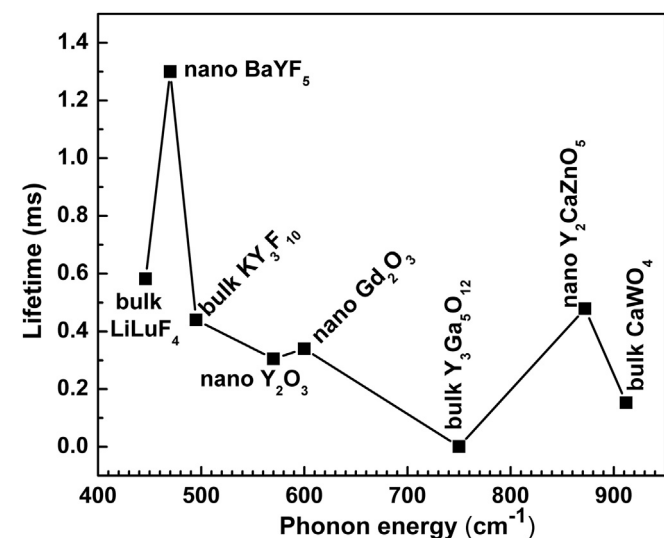


Fig. 12. Variation of lifetime of the $^4F_{9/2}$ level of 1.0 mol% of Dy^{3+} -doped crystalline materials as a function of phonon energy.

Table 3Luminescence quantum yields (Q.Y) of Dy^{3+} -doped nanophosphors.

Materials	Particle size (nm)	Preparation method	Q.Y (%)	Ref.
BYF1Dy	16	hydrothermal	4.64	Present
BYF2Dy	17	hydrothermal	11.61	Present
BYF3Dy	16	hydrothermal	11.60	Present
BYF5Dy	16	hydrothermal	10.68	Present
2Dy ³⁺ : CaMoO ₄	20	hydrothermal	5.0	[45]
2Dy ³⁺ : YVO ₄	25	Urea hydrolysis	7.0	[46]

nanocrystals is measured using an integrating sphere. The quantum yields are calculated according to the equation described elsewhere [44], where diffuse reflectance of the incident and emitted photons were collected from a powdered sample placed in an integrating sphere through a time-correlated single photon counting system. The quantum yield for these Dy^{3+} -doped BaYF₅ phosphors have been found to be increased from 4.64% to 11.61% as the concentration of Dy^{3+} increases from 1.0 mol% to 2.0 mol% and then decreased to 10.68% as concentration increased to 5.0 mol%. Similar trend has also been observed in emission intensities as function of Dy^{3+} ions concentrations (see inset of Fig. 9(a) and (b)). Table 3 shows the quantum yields for Dy^{3+} -doped BaYF₅ phosphors along with reported Dy^{3+} -doped phosphors [45,46]. The lower quantum yield of the Dy^{3+} -doped BaYF₅ nanocrystals may be due to

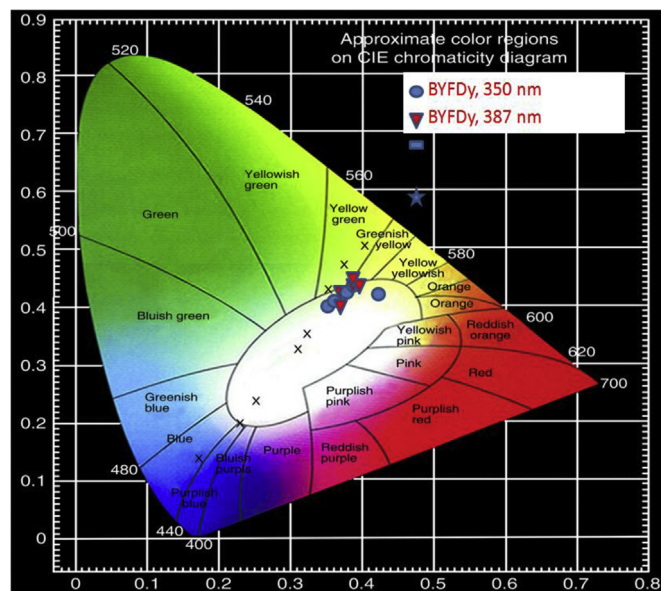


Fig. 13. Commission International d'Eclairage (CIE) color diagram. The color coordinates obtained for Dy^{3+} -doped BaYF₅ nanocrystals under 350 and 387 nm excitation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the presence of hydroxyl groups (OH^-) on the surface of the nanocrystals. Similar lower quantum yields have been observed for Dy^{3+} : CaMoO_4 [45], and Dy^{3+} : YVO_4 [46] nanophosphors.

3.6. CIE coordinates and CCT values

The CIE chromaticity coordinates of all the Dy^{3+} -doped BaYF_5 nanocrystals are determined from their luminescence spectra which are excited by 350 and 387 nm wavelengths. The detailed procedure for evaluation of CIE coordinates is described elsewhere [47]. The CIE coordinates for 350 nm excitation are found as (0.354, 0.402), (0.363, 0.415), (0.388, 0.438), (0.420, 0.425), and (0.396, 0.439) for BYF1Dy, BYF2Dy, BYF3Dy, BYF4Dy, and BYF5Dy nanocrystals, respectively (see Fig. 13, (●)). For 387 nm excitation, the CIE coordinates are found as (0.377, 0.409), (0.370, 0.421), (0.389, 0.439), (0.397, 0.439), and (0.389, 0.449) for BYF1Dy, BYF2Dy, BYF3Dy, BYF4Dy, and BYF5Dy nanocrystals, respectively (see Fig. 13, (▼)). The calculated CIE coordinates of the present materials are within the white region (see Fig. 13) but slightly far from the neutral point ($x = 0.333$, $y = 0.333$). It is also found that no significant change is observed with variation of concentration and excitation wavelength.

The luminescence quality of BaYF_5 nanocrystals is evaluated in terms of correlated color temperature (CCT). McCamy [48] has proposed the empirical formula to evaluate CCT using the color coordinates given by

$$\text{CCT} = -449n^3 + 3525n^2 - 6823n + 5520.33 \quad (6)$$

where $n = (x - x_e)/(y - y_e)$ is the inverse slope line and ($x_e = 0.332$, $y_e = 0.186$) is the epicentre.

Under 350 nm excitation, the CCT values are found as 4861,

4660, 4173, 3813, and 4012 K for BYF1Dy, BYF2Dy, BYF3Dy, BYF4Dy, and BYF5Dy nanocrystals, respectively (see Table 4). For excitation at 387 nm, the CCT values for BYF1Dy, BYF2Dy, BYF3Dy, BYF4Dy, and BYF5Dy are found to be 4283, 4507, 4156, 3992, and 4202 K, respectively (see Table 4). The obtained CCT values of the Dy^{3+} -doped BaYF_5 nanocrystals are in between to those of fluorescent tube (3935 K) and day light (5500 K) [11] and the CCT values are close to warm CCT (i.e. CCT < 5000 K) [33] compared to Dy^{3+} -doped SrF_2 [9], BaGdF_5 [10], α - NaYF_4 [24], α,β - NaYF_4 [24], β - NaYF_4 [24], $\text{Gd}_3\text{Ga}_5\text{O}_{12}$ [25], GdF_3 [26], $\text{Y}_3\text{Ga}_5\text{O}_{12}$ [32], Gd_2O_3 [35], $\text{KLa}(\text{PO}_3)_4$ [36], LuNbO_4 [49], $\text{LiSr}_4(\text{BO}_3)_3$ [50] and NaGdTlO_4 [51] phosphors. The similar results were reported for Dy^{3+} -doped $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ [11], Y_2CaZnO_5 [27], Y_2O_3 [29], CaWO_4 [31], $\text{Li}_4\text{CaB}_2\text{O}_6$ [33], Ca_2SnO_4 [52], and CdSiO_3 [53] phosphors. These results indicate that Dy^{3+} -doped BaYF_5 nanophosphors could be a potential candidate for fabrication of warm white LEDs.

4. Conclusions

Dy^{3+} -doped BaYF_5 nanocrystals have been prepared by a hydrothermal method in a single phase of the tetragonal structure with average crystallite size of around 30 nm. HRTEM results reveal that the nanocrystals have been found in a spherical shape with average about 35 nm. TG-DTA studies reveal that the synthesized nanocrystals are thermally stable up to 650 °C. The diffuse reflectance and excitation spectra confirm the incorporation of Dy^{3+} ions in BaYF_5 nanocrystals. Under 350 and 387 nm excitations, the optimized concentration of Dy^{3+} ion is found to be 2.0 mol%. The non-exponential decay curves of $^4\text{F}_{9/2}$ level are well fitted to Inokuti-Hirayama model for $S = 6$ that indicates the interaction between Dy^{3+} ions is of dipole-dipole type. The evaluated quantum yields, CIE chromaticity coordinates, and CCT values as a function of concentration indicates that the Dy^{3+} -doped BaYF_5 nanocrystals could be a suitable candidate for the generation of warm white-light.

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Table 4
CIE color coordinates (x,y) and correlated color temperatures for Dy^{3+} -doped nano and bulk crystals.

Sample	Excitation (nm)	CIE (x, y)	CCT (K)	Ref.
BYF1Dy	350	(0.354, 0.402)	4861	Present
	387	(0.377, 0.409)	4283	Present
BYF2Dy	350	(0.363, 0.415)	4660	Present
	387	(0.370, 0.421)	4507	Present
BYF3Dy	350	(0.388, 0.438)	4173	Present
	387	(0.389, 0.439)	4156	Present
BYF4Dy	350	(0.420, 0.425)	3813	Present
	387	(0.397, 0.439)	3992	Present
BYF5Dy	350	(0.386, 0.439)	4012	Present
	387	(0.389, 0.449)	4202	Present
SrF_2 : 3% Dy^{3+}	349	(0.333, 0.337)	5475	[9]
BaGdF_5 : 3% Dy^{3+}	272	(0.264, 0.303)	10764 ^a	[10]
BaGdF_5 : 3% Dy^{3+}	384	(0.253, 0.267)	15944 ^a	[10]
$\text{Lu}_3\text{Ga}_5\text{O}_{12}$: 2% Dy^{3+}	457	(0.386, 0.391)	3959	[11]
α - NaYF_4 : 2% Dy^{3+} (bulk)	350	(0.299, 0.332)	7247 ^a	[24]
α, β - NaYF_4 : 2% Dy^{3+} (bulk)	351	(0.313, 0.359)	6312 ^a	[24]
β - NaYF_4 : 2% Dy^{3+} (bulk)	351	(0.323, 0.371)	5860 ^a	[24]
$\text{Gd}_3\text{Ga}_5\text{O}_{12}$: 2% Dy^{3+}	277	(0.32, 0.34)	6073 ^a	[25]
GdF_3 : 2% Dy^{3+}	386	(0.257, 0.276)	13913 ^a	[26]
Y_2CaZnO_5 : 1% Dy^{3+}	424	(0.374, 0.440)	4495	[27]
Y_2O_3 : 1% Dy^{3+}	351	(0.41, 0.45)	3800 ^a	[29]
CaWO_4 : 1% Dy^{3+} (bulk)	353	(0.363, 0.407)	4631 ^a	[31]
$\text{Y}_3\text{Ga}_5\text{O}_{12}$: 2% Dy^{3+} (bulk)	366	(0.326, 0.326)	5816	[32]
$\text{Li}_4\text{CaB}_2\text{O}_6$: 1% Dy^{3+}	UV GaN	(0.39, 0.41)	3983	[33]
Gd_2O_3 : 1% Dy^{3+}	238	(0.36, 0.38)	5515 ^a	[35]
$\text{KLa}(\text{PO}_3)_4$: 1% Dy^{3+} (bulk)	325	(0.235, 0.229)	44003 ^a	[36]
LuNbO_4 : 1% Dy^{3+} (bulk)	261	(0.336, 0.311)	5305	[49]
$\text{LiSr}_4(\text{BO}_3)_3$: 2% Dy^{3+} (bulk)	350	(0.269, 0.303)	10286 ^a	[50]
NaGdTlO_4 : 3% Dy^{3+} (bulk)	281	(0.334, 0.353)	5439 ^a	[51]
Ca_2SnO_4 : 2% Dy^{3+} (bulk)	350	(0.431, 0.456)	3470 ^a	[52]
CdSiO_3 : 5% Dy^{3+} (bulk)	254	(0.387, 0.376)	4000	[53]

^a CCT values are calculated from the reported CIE coordinates.

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